From Streams to Review:
Analyzing Consumer Feedback on Spotify

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Table of Contents

01. Introduction

02. Sentiment Analysis

03. Comparative Analysis

04. Topic Modeling Analysis

05. Classification

06. Conclusions and Implications

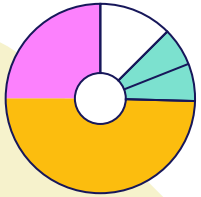
01.

Introduction

Research Question

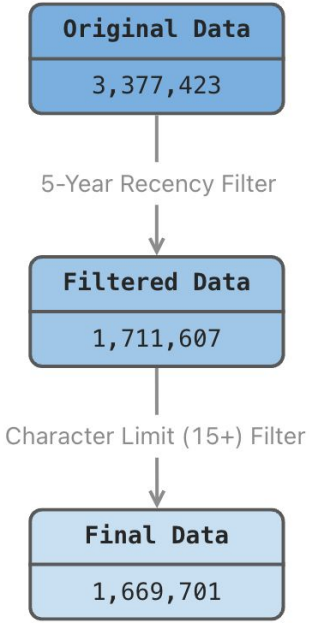
- **We developed our research questions by creating three sub-questions to zoom in and conduct a deep investigation.**

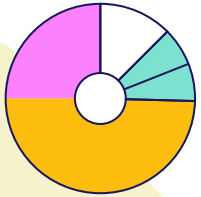
- **Research Question:** How can Spotify Play Store reviews be analyzed to understand evolving user satisfaction, brand perception, and review patterns?
  - **Sub-1:** What is the overall sentiment trend of Spotify reviews?
  - **Sub-2:** What are the most common words and topics in user reviews?
  - **Sub-3:** Can review ratings be predicted based on review text features?



Data Source and Filtering

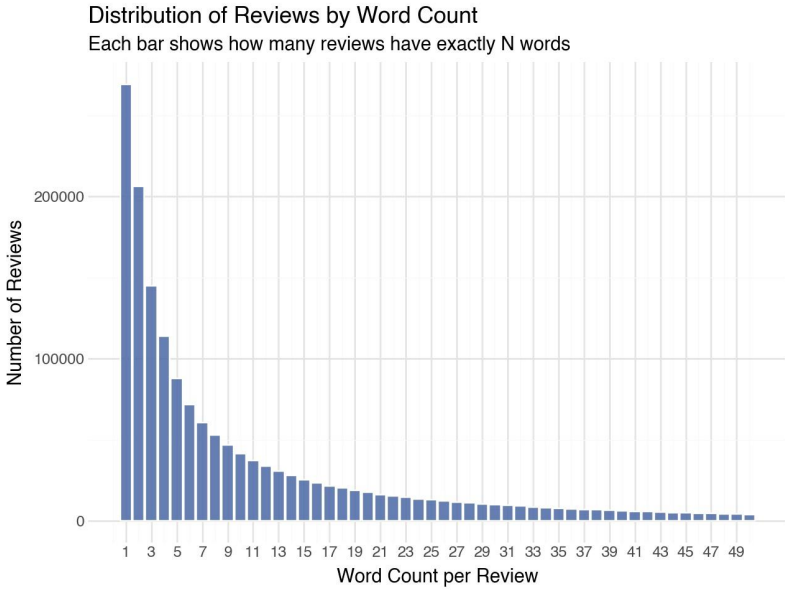
- **Dataset name:** Spotify Google Store Review (From Kaggle)
- **Number of Reviews:** 3377423
- **Variable Description:**
  - *review id*
  - *author id*
  - *author name*
  - *review text*
  - *review ratings (from 1 to 5)*
  - *review likes*
  - *author app version*
  - *review timestamp*

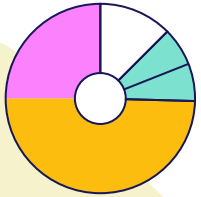




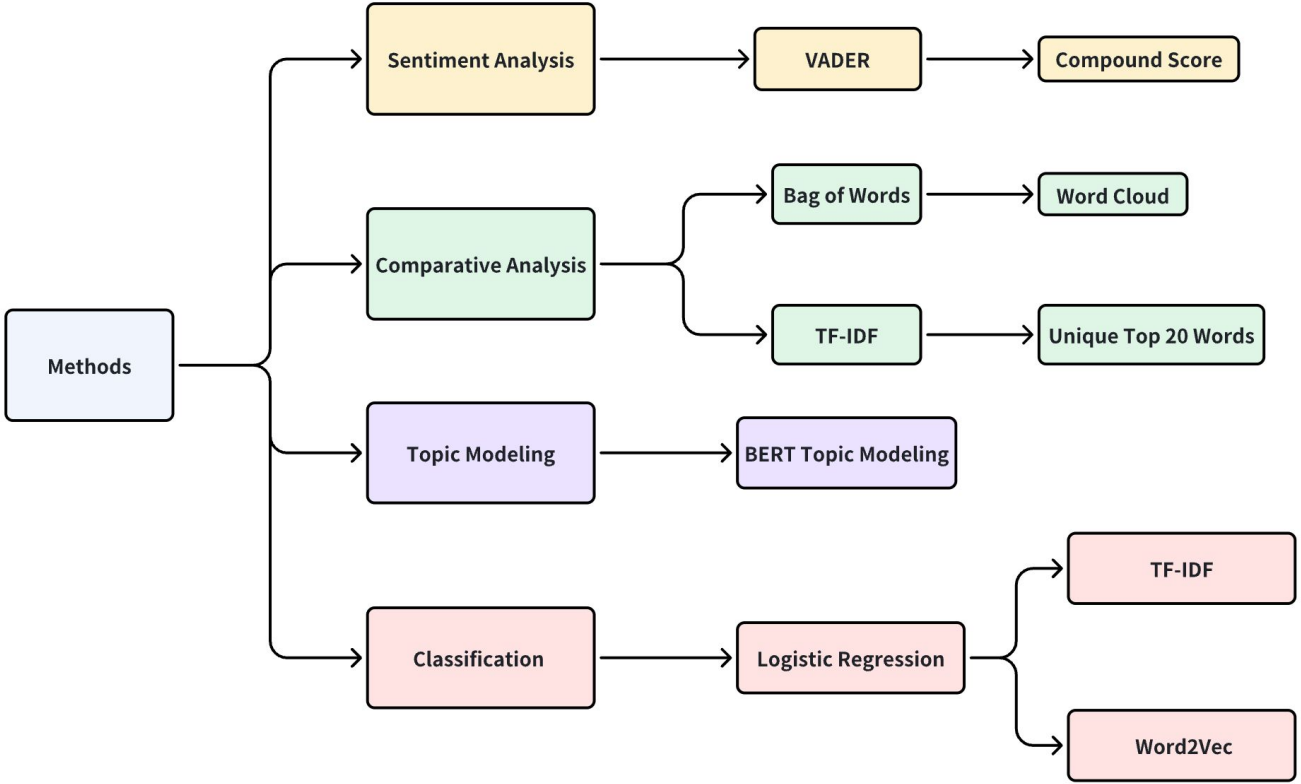
Data Transformation: Review Length Categorization

Word Count	Type 1	Type 2
1 - 3	Very Short	Short
4 - 6	Short	
7 - 10	Medium	Long
10 +	Long	





Methods


Methods
Sentiment Analysis
VADER
Compound Score
Comparative Analysis
Bag of Words
Word Cloud
TF-IDF
Unique Top 20 Words
Topic Modeling
BERT Topic Modeling
Classification
Logistic Regression
TF-IDF
Word2Vec

02.

Sentiment Analysis

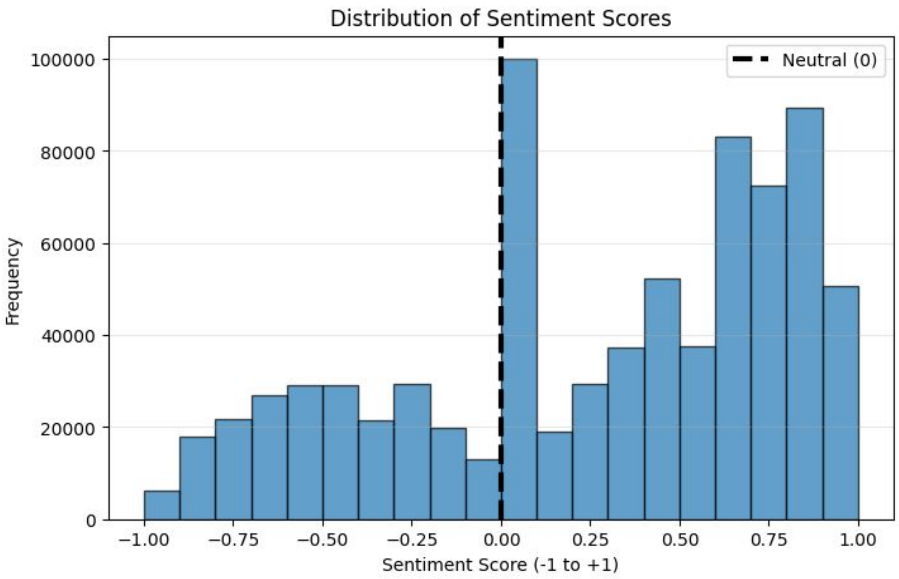
Sentiment Score Distribution

Positively Skewed

- The largest cluster is near-neutral (0).
- The second-largest group is strongly positive (0.6-0.9).

Negatives are Scarce

- The entire negative range is significantly smaller than the positive range.
- 75% of all scores are -0.12 or higher.



Sentiment Score vs. Review Rating

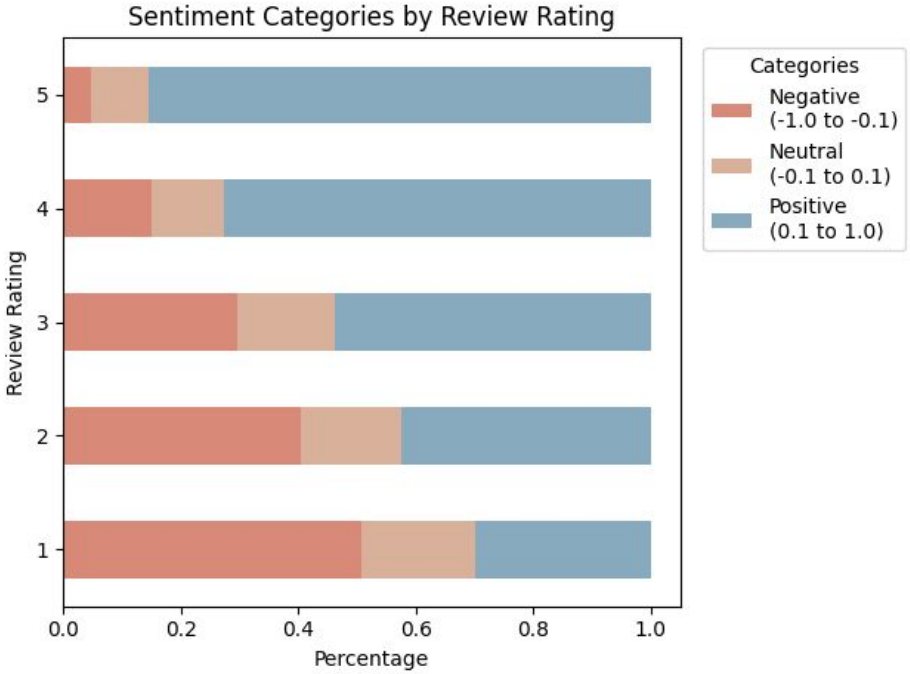
Strong Correlation

- As the user's Review Rating increases, the Sentiment Score also clearly increases.

The 5-Star Anomaly

- A large group of 5-star reviews contains text with highly negative sentiment scores.
- Explanation: people notes their frustrations but still give 5 points-"Its really good tbh, the ads get a little annoying....."

Sentiment Category vs. Review Rating



Progressive Trend

- The share of negative text reviews consistently shrinks as the rating increases from 1 to 5.

Mixed Negative Signals

- 1-star reviews is highly conflicted, containing only ~50% negative text reviews, but 30% of reviews with positive sentiment score.

03.

Comparative Analysis

Word Cloud by Review Rating

reviews with > = 7 words, removed stop words

Review = 1

TOP 5:Premium, ads,update,playlist,fix

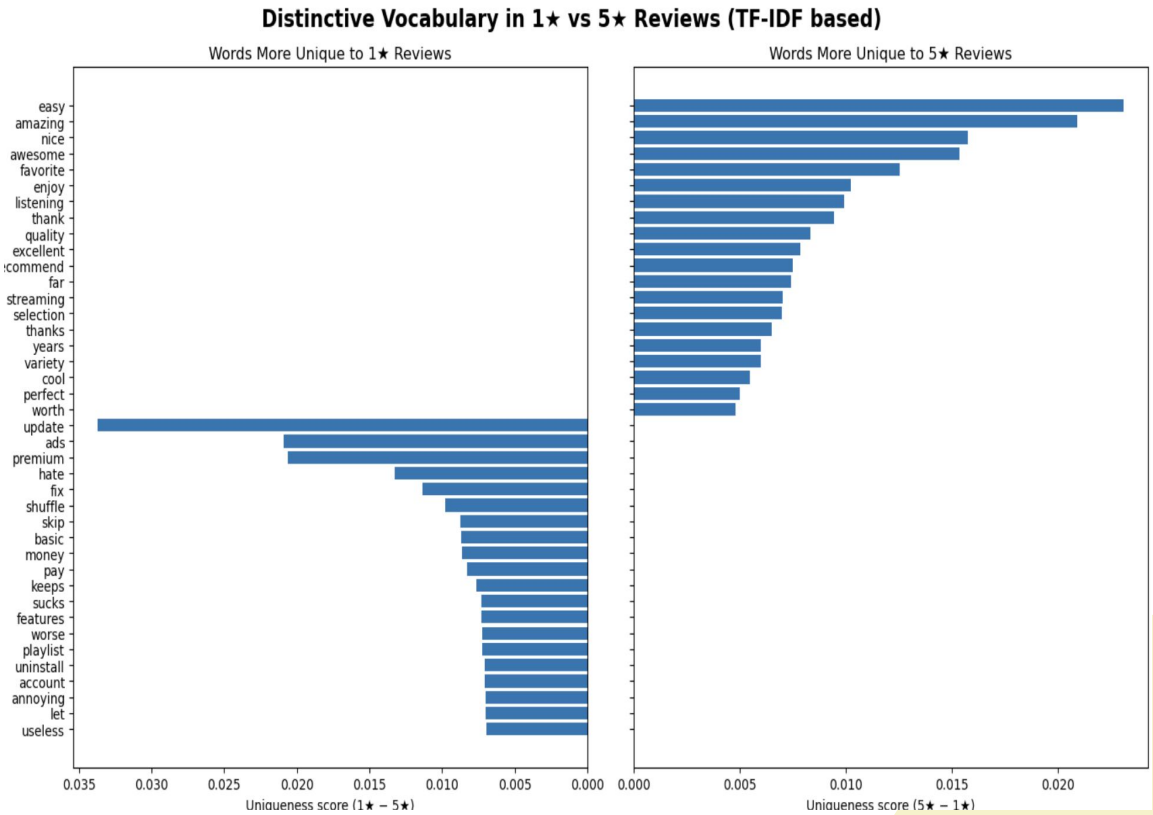
Review = 5

TOP 5:Premium, Easy, amazing, listening,ads

TF-IDF comparison of good reviews and bad reviews

- 1-star comments focus heavily on **functional pain points and negative terms, like fix, skip**

- 5-star reviews highlight **positive experiences and usability**, using words like easy, amazing, and awesome.



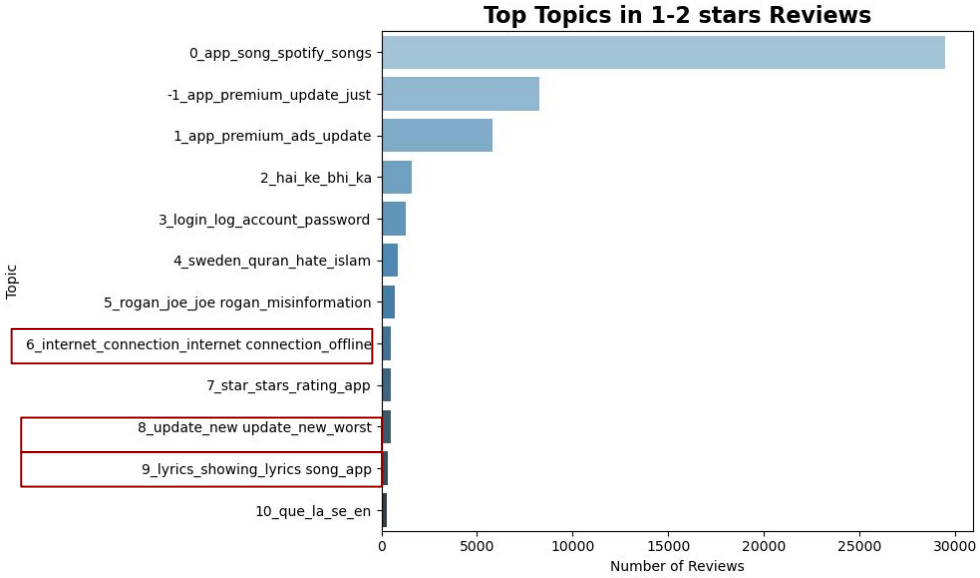
04.

Topic Modeling

Topic Modeling

Methodology

- Using bert topic modeling for 3 models on 3 different subsets of the data
- 1-2 star reviews, 3 star reviews, and 4-5 star reviews

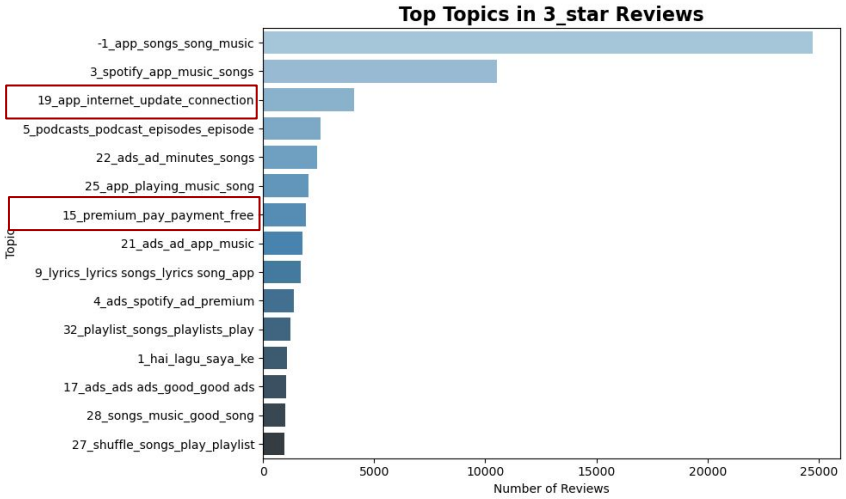


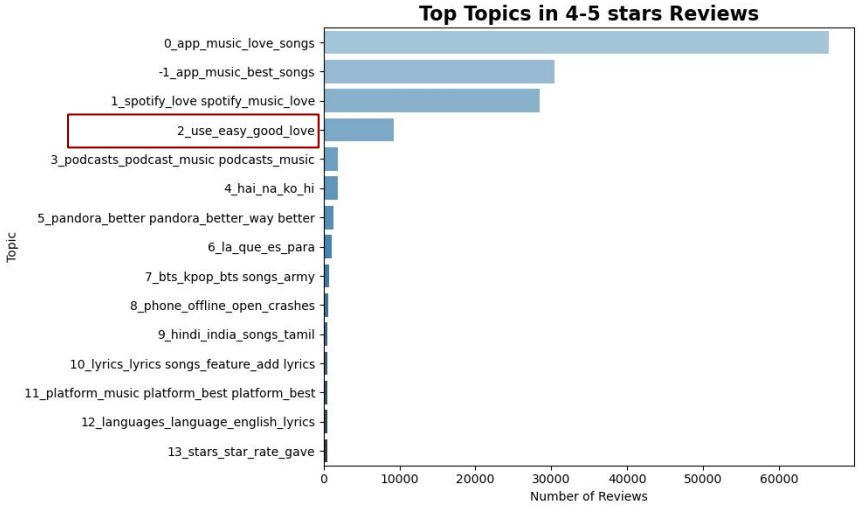
Representative Doc

- “App did not update for better option”
- “could not skip song”
- “account setting issue with password”

Topic Modeling

The top two to three topics are consistent across all three models, but their content shifts by rating: 1–2 star reviews emphasize **negative issues**, 3-star reviews focus on **suggested improvements**, and 4–5 star reviews highlight **positive experiences and appreciation**





Representative Doc

- “Good app but cannot download song”
- “Use for long time but took too long to load”
- “Love it but no ads need premium”

Representative Doc

- “Best music app ever”
- “Love the music and context”

05.

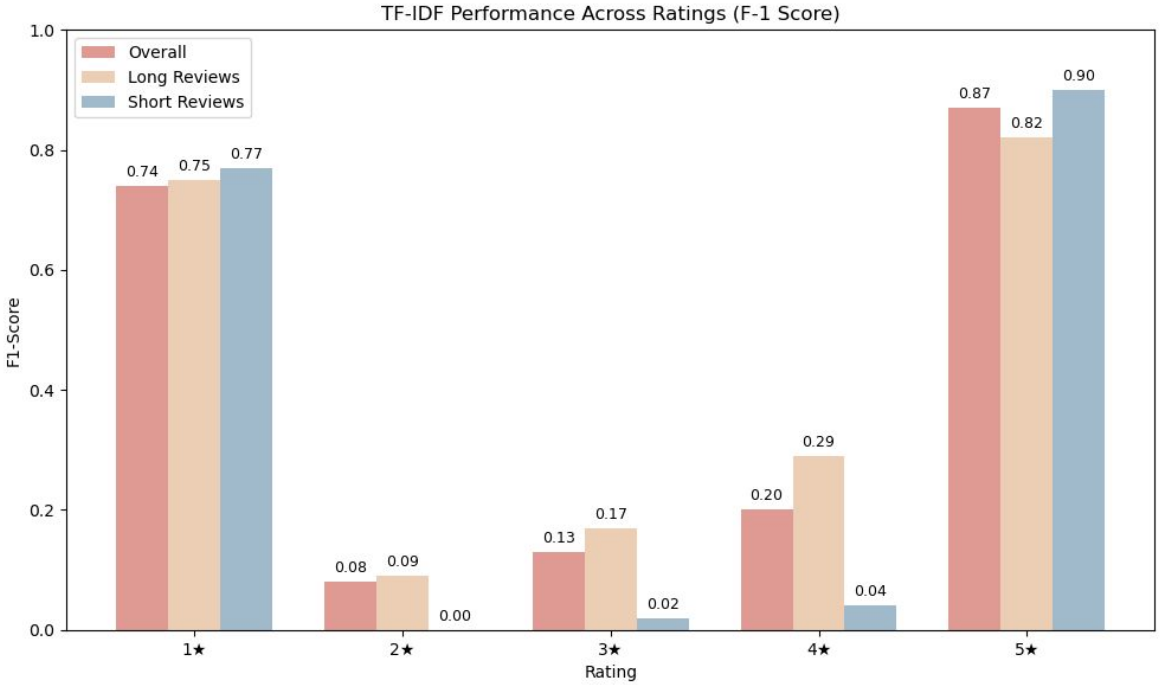
Classification

Comparison between TF-IDF and Word2Vec

Category	Accuracy (TF-IDF)	F1-Score (TF-IDF)	Accuracy (Word2Vec)	F1-Score (Word2Vec)
Overall	0.75	0.70	0.71	0.64
Long text	0.67	0.62	0.63	0.56
Short text	0.83	0.77	0.80	0.75

- **TF-IDF outperforms Word2Vec across all categories**
  - Both Accuracy and weighted F-1 Score are consistently higher for TF-IDF
- **Short text performs better than long text for both TF-IDF and Word2Vec**
  - Short text contain very dense sentiment words, so both methods classify them more reliably.
  - Longer text is noisier, with mixed sentiment clauses, making logistic regression less effective.
- Best-performing model: TF-IDF on short text (Accuracy 0.83, F-1 0.77)
- Worst-performing model: Word2Vec on long text (Accuracy 0.60, F-1 0.52)

TF-IDF Performance Across Ratings (F-1 Score)



- **TF-IDF predicts extreme ratings (1 & 5) very well regardless of text length**
  - These have the most distinct language (strong praise or strong complaints).
- **TF-IDF performs very poor on mid-level ratings (2-4)**
  - These have vague or mixed language which TF-IDF cannot uniquely associate with one rating.
- The overall F-1 Score is inflated by strong performance on extreme ratings, while mid-level ratings remain poorly predicted.

06.

Conclusions & Implications

Conclusion & Implications

Sentiment Analysis

Sentiment Distribution: Positively Skewed
Strong Correlation: As user's review rating increases, sentiment score also increases.

Topic Modeling

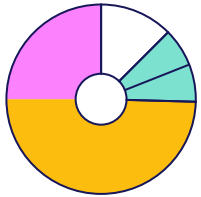
Sentiment shift: same topics or themes, but tone shifts for different ratings.
Different rating focus: low rating (problems); mid-rating (fixes); high rating (positive experience)

Comparative Analysis

Word Cloud: Distinct top 5 words for both extremes
Different Focus: 1-star (functional pain points); 5-star (positive experience)

Classification

Model Comparison: TF-IDF > Word2Vec; short review > long review
F-1 Inflation: perform well on extremes (1 & 5- star); inflated overall F-1 Score



Thanks!